

ELECTIVE COURSE-2.2

URA621: INDUSTRIAL ROBOTICS				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: Students explore robotics systems and the role they play in industry, actively designing a complete robotics systems with a robot arm, mobile robot and peripheral tools to stimulate manufacturing process.				
Syllabus Introduction to Robotics: The Term robotics, The history of robotics, Human-robot collaboration, Humanoids, service robots, mobile robot platform, space robots, Autonomous vehicles. Industrial Robots: Introduction to industrial robots, definition of industrial robots, Handling devices, kinematic representation, load characteristics, geometric characteristics, kinematic characteristics, accuracy characteristics, Robot types: Articulated arm robots, SCARA robot, gantry robot, delta robot, manufacturers of robot system. Areas of application and system components: Application of Industrial robots, production engineering, surface machining technology, connection technology, separating technology, robots in automation pyramid, system components of an industrial robot, actuator systems in robotics, Axis control, end effectors, sensors. Robot controller: Introduction to robot controller, control panel of a robot controller, handheld programming device of a robot controller, geometrical kinematics, Robot coordinate systems, motion controller. Introduction to collaborative robotics (HRC): Introduction to human-robot collaboration, Level of cooperation between human and robot, workspace of an HRC system, cooperation and guidelines for human-robot collaboration: HRC systems and their requirements, contactless cooperation, collaborative cooperation, commissioning of HRC system, Intuitive programming in human-robot collaboration, programming industrial robots, intuitive programming, Cobotta as collaborative robot. Autonomous Mobile robot: Introduction to Robotino (Mobile robot), areas of applications				

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of mobile robots, tasks in industry automated guided vehicle system, the learning system Robotino, travel behavior of Mobile robot, functional blocks of mobile robot Collaborative Robot, odometry in mobile robot, trigonometry and vector calculus to travel a defined distance in a multi-axis system using the motor controller, Marker Identification, Environment Mapping in mobile robot, closed-loop control structure for the path tracking, integration of Mobile robot with Collaborative Robot,

Robot with Assembly operation: Industrial Robot: Gantry robots, SCARA robots, vertical articulated Arm robots, system components of industrial robot, characteristics of industrial robot, Grippers: Mechanical grippers, suction grippers, Tool-changing systems, robot controllers: cartesian coordinate systems, robot coordinate systems, control of robot movements, sensors: Inductive proximity switches, optical proximity switches, camera systems, programming industrial robots: online procedure, offline procedure, collision avoidance, safety measures, 6 Axis Robot application, Cylinder manufacturing operation.

Laboratory Work

1. Incoming inspection and commissioning of Robotino
2. Linear travel and positioning of Robotino
3. Path tracking of a driverless transport system with two diffusion sensor
4. Accurately positioned approach to a loading station
5. Approaching the obstacle and maintaining specified distance
6. Path tracking a driverless transport system using an analog inductive sensor
7. Creating an environment map and navigating the Robotino
8. Approaching the markers autonomously
9. Path tracking an FTS using a camera
10. Commissioning and integration of cobotta with Robotino
11. Programming the Industrial Robot to perform assembly operation
12. Programming the Industrial Robot with TCP
13. Programming the robot with Cartesian and other programming formats, etc.

Course Learning Outcomes (CLOs)

The students will be able to:

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